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RULE · Linear vs. exponential (table test)

Common DIFFERENCE constant → linear. Common RATIO constant → exponential.

Type	Values	Test
Linear	3, 7, 11, 15	difference always +4
Exponential	5, 10, 20, 40	ratio always ×2

RULE · Growth / decay formula

$y = a(1 + r)^x$ (growth) $\qquad y = a(1 - r)^x$ (decay)

$a = 200, r = 5\%, x = 4$ yr: $\quad y = 200(1.05)^4 \approx 243$

r is a decimal: $7\% = 0.07$, not 1.7.

RULE · Half-life

$$y = a\left(\frac{1}{2}\right)^{t/h}$$

$a = 160$ g, $h = 4$ yr, $t = 12$ yr: $\quad y = 160\left(\frac{1}{2}\right)^{12/4} = 160\left(\frac{1}{2}\right)^3 = 20$ g

Each half-life multiplies the remaining amount by $\frac{1}{2}$ – it does not subtract a fixed amount.